

Definitions in **Sarason/** and **R2.lean**

Lean Formalization Project

September 10, 2026

This document provides a comprehensive list of all definitions manually formalized in the **ComplexAnalysis** project, specifically within **R2.lean** and the **Sarason/** directory. For each definition, the raw Lean code is presented first, followed by its mathematical interpretation.

1 **ComplexAnalysis/R2.lean**

- **sqNorm** ($x \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The squared Euclidean norm.

```
noncomputable def sqNorm (x : ℝ × ℝ) : ℝ := x.1^2 + x.2^2
```

$$\text{sqNorm}(x) = x_1^2 + x_2^2$$

- **sqDist** ($x, y \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The squared Euclidean distance.

```
noncomputable def sqDist (x y : ℝ × ℝ) : ℝ :=  
  (x.1 - y.1)^2 + (x.2 - y.2)^2
```

$$\text{sqDist}(x, y) = \text{sqNorm}(x - y)$$

- **euclideanNorm** ($x \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The standard Euclidean norm.

```
noncomputable def euclideanNorm (x : ℝ × ℝ) : ℝ :=  
  Real.sqrt (sqNorm x)
```

$$\text{euclideanNorm}(x) = \sqrt{x_1^2 + x_2^2}$$

- **euclideanDist** ($x, y \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The Euclidean distance.

```
noncomputable def euclideanDist (x y : ℝ × ℝ) : ℝ :=  
  Real.sqrt (sqDist x y)
```

$$\text{euclideanDist}(x, y) = \sqrt{\text{sqDist}(x, y)}$$

- **LimitRtoR2** ($f : \mathbb{R} \rightarrow \mathbb{R}^2, a \in \mathbb{R}, L \in \mathbb{R}^2$):

```
def LimitRtoR2 (f : ℝ → ℝ × ℝ) (a : ℝ) (L : ℝ × ℝ) : Prop :=  
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ, 0 < |x - a| ∧ |x - a| < δ → euclideanDist (f x) L  
    < ε
```

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0, \forall x, 0 < |x - a| < \delta \implies \text{euclideanDist}(f(x), L) < \epsilon$$

- **LimitR2toR** ($f : \mathbb{R}^2 \rightarrow \mathbb{R}, a \in \mathbb{R}^2, L \in \mathbb{R}$):

```
def LimitR2toR (f : ℝ × ℝ → ℝ) (a : ℝ × ℝ) (L : ℝ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ × ℝ, 0 < euclideanDist x a ∧ euclideanDist x a < δ
    → |f x - L| < ε
```

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}^2, 0 < \text{euclideanDist}(x, a) < \delta \implies |f(x) - L| < \epsilon$$

- LimitR2toR2 (f : ℝ² → ℝ², a ∈ ℝ², L ∈ ℝ²):

```
def LimitR2toR2 (f : ℝ × ℝ → ℝ × ℝ) (a : ℝ × ℝ) (L : ℝ × ℝ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ × ℝ, 0 < euclideanDist x a ∧ euclideanDist x a < δ
    → euclideanDist (f x) L < ε
```

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}^2, 0 < \text{euclideanDist}(x, a) < \delta \implies \text{euclideanDist}(f(x), L) < \epsilon$$

- ConvergesR2 (seq : ℕ → ℝ², L ∈ ℝ²):

```
def ConvergesR2 (seq : ℕ → ℝ × ℝ) (L : ℝ × ℝ) : Prop :=
  ∀ ε > 0, ∃ N : ℕ, ∀ n ≥ N, euclideanDist (seq n) L < ε
```

$$\lim_{n \rightarrow \infty} \text{seq}(n) = L \iff \forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \text{euclideanDist}(\text{seq}(n), L) < \epsilon$$

- HasFDerivAt_R2_eps (u, v : ℝ² → ℝ, u_{x0}, u_{y0}, v_{x0}, v_{y0} ∈ ℝ, a ∈ ℝ²): Defines the Fréchet derivative of F(x, y) = (u(x, y), v(x, y)) at a. Let L(h) = $\begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} h$.

```
def HasFDerivAt_R2_eps (u v : ℝ × ℝ → ℝ) (ux0 uy0 vx0 vy0 : ℝ) (a : ℝ × ℝ) :
  Prop :=
  LimitR2toR (fun h =>
    euclideanNorm (
      (u (a.1 + h.1, a.2 + h.2) - u a) - (ux0 * h.1 + uy0 * h.2),
      (v (a.1 + h.1, a.2 + h.2) - v a) - (vx0 * h.1 + vy0 * h.2)
    ) / euclideanNorm h
  ) (0, 0) 0
```

$$\lim_{h \rightarrow 0} \frac{\left\| F(a+h) - F(a) - \begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} h \right\|}{\|h\|} = 0$$

$$\lim_{(h_1, h_2) \rightarrow (0,0)} \frac{\left\| \begin{pmatrix} u(a_1 + h_1, a_2 + h_2) - u(a_1, a_2) \\ v(a_1 + h_1, a_2 + h_2) - v(a_1, a_2) \end{pmatrix} - \begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right\|}{\|(h_1, h_2)\|} = 0$$

- DifferentiableAt_R2_eps (u, v : ℝ² → ℝ, a ∈ ℝ²):

```
def DifferentiableAt_R2_eps (u v : ℝ × ℝ → ℝ) (a : ℝ × ℝ) : Prop :=
  ∃ (ux0 uy0 vx0 vy0 : ℝ), HasFDerivAt_R2_eps u v ux0 uy0 vx0 vy0 a
```

$$\exists u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}, \text{HasFDerivAt_R2_eps}(u, v, u_{x0}, u_{y0}, v_{x0}, v_{y0}, a)$$

- HasDerivAt_RtoR2_eps (f : ℝ → ℝ², f'₀ ∈ ℝ², t₀ ∈ ℝ): Defines the derivative of a parametric curve in ℝ².

```
def HasDerivAt_RtoR2_eps (f : ℝ → ℝ × ℝ) (f'₀ : ℝ × ℝ) (t₀ : ℝ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ t : ℝ, 0 < |t - t₀| ∧ |t - t₀| < δ →
    euclideanDist (f t) (f t₀ + (f'₀.1 * (t - t₀), f'₀.2 * (t - t₀))) < ε * |t - t₀|
```

$$\lim_{t \rightarrow t_0} (f(t) - f(t_0) - (t - t_0)f'_0) = 0$$

2 ComplexAnalysis/Sarason/Definitions.lean

- HasLimitAt_eps ($f : \mathbb{C} \rightarrow \mathbb{C}, L \in \mathbb{C}, z_0 \in \mathbb{C}$):

```
def HasLimitAt_eps (f : ℂ → ℂ) (L : ℂ) (z₀ : ℂ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ z, 0 < ‖z - z₀‖ ∧ ‖z - z₀‖ < δ → ‖f z - L‖ < ε

-- Weierstrass definition of a limit at infinity.
```

$$\lim_{z \rightarrow z_0} f(z) = L$$

- HasLimitAtInf_eps ($f : \mathbb{C} \rightarrow \mathbb{R}, L \in \mathbb{R}$):

```
def HasLimitAtInf_eps (f : ℂ → ℝ) (L : ℝ) : Prop :=
  ∀ ε > 0, ∃ M > 0, ∀ z, ‖z‖ > M → |f z - L| < ε

-- Weierstrass definition of having a complex derivative at z₀.
```

$$\lim_{|z| \rightarrow \infty} f(z) = L$$

- HasDerivAt_eps ($f : \mathbb{C} \rightarrow \mathbb{C}, f'_0 \in \mathbb{C}, z_0 \in \mathbb{C}$): Complex differentiability at a point.

```
def HasDerivAt_eps (f : ℂ → ℂ) (f'₀ : ℂ) (z₀ : ℂ) : Prop :=
  HasLimitAt_eps (fun z => (f z - f z₀) / (z - z₀)) f'₀ z₀

-- Weierstrass definition of complex differentiability at z₀.
```

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'_0$$

- DifferentiableAt_eps ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$):

```
def DifferentiableAt_eps (f : ℂ → ℂ) (z₀ : ℂ) : Prop :=
  ∃ f'₀, HasDerivAt_eps f f'₀ z₀

-- Weierstrass definition of partial derivative with respect to x.
```

f is differentiable at z_0

- HasPartialDerivX_C_to_R_eps / HasPartialDerivY_C_to_R_eps: Partial derivatives of a real-valued function of a complex variable (identifying $\mathbb{C} \cong \mathbb{R}^2$).

```
def HasPartialDerivX_C_to_R_eps (u : ℂ → ℝ) (ux₀ : ℝ) (z₀ : ℂ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ, 0 < |x - z₀.re| ∧ |x - z₀.re| < δ →
    |(u ((x : ℂ) + (z₀.im : ℂ) * I) - u z₀) / (x - z₀.re) - ux₀| < ε

-- Weierstrass definition of partial derivative with respect to y.
```

$$\frac{\partial u}{\partial x}(z_0) = u_x$$

- HasPartialDerivY_C_to_R_eps

```
def HasPartialDerivY_C_to_R_eps (u : ℂ → ℝ) (uy₀ : ℝ) (z₀ : ℂ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ y : ℝ, 0 < |y - z₀.im| ∧ |y - z₀.im| < δ →
    |(u ((z₀.re : ℂ) + (y : ℂ) * I) - u z₀) / (y - z₀.im) - uy₀| < ε

-- Equivalence theorem between Mathlib's limit at a point (using punctured
  neighborhoods) and Weierstrass epsilon-delta limit.
```

$$\frac{\partial u}{\partial y}(z_0) = u_y$$

- `deriv_eps` ($f : \mathbb{C} \rightarrow \mathbb{C}$, $z_0 \in \mathbb{C}$, $h : \text{DifferentiableAt_eps } f \ z_0$): The classical derivative value $f'(z_0)$ extracted via choice.

```
noncomputable def deriv_eps (f : ℂ → ℂ) (z₀ : ℂ) (h : DifferentiableAt_eps f
  z₀) : ℂ :=
  Classical.choose h
```

$$\text{deriv_eps}(f, z_0) = f'(z_0)$$

- `HasDerivAt_R_to_C_eps` ($f : \mathbb{R} \rightarrow \mathbb{C}$, $f'_0 \in \mathbb{C}$, $t_0 \in \mathbb{R}$): Derivative of a complex-valued path parameterized by a real variable.

```
def HasDerivAt_R_to_C_eps (f : ℝ → ℂ) (f'₀ : ℂ) (t₀ : ℝ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ t : ℝ, 0 < |t - t₀| ∧ |t - t₀| < δ → ||(f t - f t₀) / ((t
    - t₀ : ℝ) : ℂ) - f'₀|| < ε
```

$$\lim_{t \rightarrow t_0} \frac{f(t) - f(t_0)}{t - t_0} = f'_0$$

- `HolomorphicAt_eps` ($f : \mathbb{C} \rightarrow \mathbb{C}$, $z_0 \in \mathbb{C}$): f is complex-differentiable in some open neighborhood around z_0 .

```
def HolomorphicAt_eps (f : ℂ → ℂ) (z₀ : ℂ) : Prop :=
  DifferentiableAt_eps f z₀

/-- A function is holomorphic on a set if it is complex differentiable at
  every point of the set. -/
```

f is holomorphic at $z_0 \iff f$ is complex-differentiable on some open ball around z_0

- `HolomorphicOn_eps` ($f : \mathbb{C} \rightarrow \mathbb{C}$, $G \subset \mathbb{C}$): f is holomorphic at every point in the open set G .

```
def HolomorphicOn_eps (f : ℂ → ℂ) (G : Set ℂ) : Prop :=
  ∀ z ∈ G, DifferentiableAt_eps f z

end Sarason

/-!
  NOTE FOR FUTURE AGENTS:
  Currently, some theorems in 'Chapter2.lean' (such as '
    conformal_implies_holomorphic')
  use Mathlib's 'DifferentiableAt ℝ f z' to represent continuous first
  partial derivatives / real-differentiability.
  Eventually, we want to replace this with our own custom epsilon-delta
  definitions
  (similar to 'HasDerivAt_eps') that will define ℝ² differentiability from
  scratch.
-/

-- ℝ² Euclidean Metrics (from AnalysTSP)
```

f is holomorphic on G

3 ComplexAnalysis/Sarason/Chapter1.lean

- `chordalMetric` ($z, w \in \mathbb{C} \rightarrow \mathbb{R}$): The chordal distance between two points in the complex plane mapped to the Riemann sphere.

```
noncomputable def chordalMetric (z w : ℂ) : ℝ :=
  (2 * Complex.abs (z - w)) / (Real.sqrt (1 + Complex.normSq z) * Real.sqrt (1
    + Complex.normSq w))
```

$$\chi(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}$$

- `chordalMetricToInf` ($z \in \mathbb{C} \rightarrow \mathbb{R}$): The chordal distance from a point z to the point at infinity ∞ .

```
noncomputable def chordalMetricToInf (z : ℂ) : ℝ :=
  2 / Real.sqrt (1 + Complex.normSq z)
```

$$\chi(z, \infty) = \frac{2}{\sqrt{1 + |z|^2}}$$

- `Φ` (Stereographic Projection)

```
noncomputable def Φ (z : ℂ) : EuclideanSpace ℝ (Fin 3) :=
  ![ (2 * z.re) / (1 + Complex.normSq z),
    (2 * z.im) / (1 + Complex.normSq z),
    (Complex.normSq z - 1) / (1 + Complex.normSq z) ]
```

$$\Phi(z) = \left(\frac{2\operatorname{Re}(z)}{1 + |z|^2}, \frac{2\operatorname{Im}(z)}{1 + |z|^2}, \frac{|z|^2 - 1}{1 + |z|^2} \right) \in S^2$$

- `ChordalC`: Type alias for $\mathbb{C}$ equipped with the chordal metric.

```
def ChordalC : Type := ℂ
```

ChordalC = $\mathbb{C}$ (equipped with chordal metric)

4 ComplexAnalysis/Sarason/Chapter2.lean

- `del` (∂)

```
def del (f : ℂ → ℂ) (z : ℂ) : ℂ :=
  let L := fderiv ℝ f z
  let df_dx := L 1
  let df_dy := L I
  (1 / 2 : ℂ) * (df_dx - I * df_dy)
```

```
-- The  $\bar{\partial}$  operator (del-bar) for a function f : ℂ → ℂ at point z. -/
```

$$\partial f = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right)$$

- `delBar` ($\bar{\partial}$)

```

def delBar (f : ℂ → ℂ) (z : ℂ) : ℂ :=
  let L := fderiv ℝ f z
  let df_dx := L 1
  let df_dy := L I
  (1 / 2 : ℂ) * (df_dx + I * df_dy)

-- Local notation for Sarason's flavor
local notation f "z" => del f
local notation f "zbar" => delBar f

/--
  Relating Sarason's Weierstrass derivative to Mathlib's Filter-based
  HasDerivAt.
-/

```

$$\bar{\partial}f = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right)$$

- `del_eps / delBar_eps` ($u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}$): Wirtinger derivatives expressed purely in terms of the real partial derivatives of $f = u + iv$.

```

noncomputable def del_eps (ux uy vx vy : ℝ) : ℂ :=
  (1 / 2 : ℂ) * ( (ux + vy) + I * (vx - uy) )

```

$$\partial f = \frac{1}{2}(u_{x0} + v_{y0}) + \frac{i}{2}(v_{x0} - u_{y0})$$

- `delBar_eps`

```

noncomputable def delBar_eps (ux uy vx vy : ℝ) : ℂ :=
  (1 / 2 : ℂ) * ( (ux - vy) + I * (vx + uy) )

```

$$\bar{\partial}f = \frac{1}{2}(u_{x0} - v_{y0}) + \frac{i}{2}(v_{x0} + u_{y0})$$

- `path_in_C` ($x, y \in \mathbb{C}$): A continuous path $\gamma : [0, 1] \rightarrow \mathbb{C}$ starting at x and ending at y .

```

def path_in_C (x y : ℂ) : Type := Path x y

/-- Velocity / derivative of a path  $\gamma : \text{path\_in\_C } x \ y$  at parameter time  $t \in \mathbb{R}$ 
  . -/

```

$$\gamma : [0, 1] \rightarrow \mathbb{C} \text{ where } \gamma(0) = x, \gamma(1) = y$$

- `translated_path` ($c, x, y \in \mathbb{C}, \gamma : \text{path_in_C } x \ y$): A path shifted by the constant c , creating a path from $c + x$ to $c + y$.

```

noncomputable def translated_path {x y : ℂ} (c : ℂ) (γ : path_in_C x y) :
  path_in_C (c + x) (c + y) where
  toFun := fun t => c + γ.toFun t
  continuous_toFun := Continuous.add continuous_const γ.continuous_toFun
  source' := by { change c + γ.toFun 0 = c + x; have h : γ.toFun 0 = x := (γ :
    Path x y).source'; rw [h] }
  target' := by { change c + γ.toFun 1 = c + y; have h : γ.toFun 1 = y := (γ :
    Path x y).target'; rw [h] }

```

$$\gamma_c(t) = c + \gamma(t)$$

- `pathDeriv` ($\gamma : \text{path_in_C } x \ y, t \in \mathbb{R}$): The tangent vector $\gamma'(t) \in \mathbb{C}$, extracted via choice from `HasDerivAt_R_to_C_eps` extended to $\mathbb{R}$.

```
noncomputable def pathDeriv {x y : ℂ} (γ : path_in_C x y) (t : ℝ) : ℂ :=
  deriv γ.extend t
```

```
/-- The direction of a path γ at time t is the argument of its derivative. -/
```

$$\gamma'(t) = \frac{d\gamma}{dt}$$

- `pathDirection` ($\gamma : \text{path_in_C } x \ y, t \in \mathbb{R}$): The argument (angle) of the tangent vector, $\arg(\gamma'(t))$.

```
noncomputable def pathDirection {x y : ℂ} (γ : path_in_C x y) (t : ℝ) : ℝ :=
  arg (pathDeriv γ t)
```

```
/-- The angle between two paths γ₁ and γ₂ at times t₁ and t₂ is the difference
of their directions. -/
```

$$\text{Direction}(\gamma, t) = \arg(\gamma'(t))$$

- `pathAngle` ($\gamma_1, t_1, \gamma_2, t_2$): The angle between two intersecting paths.

```
noncomputable def pathAngle {x₁ y₁ x₂ y₂ : ℂ} (γ₁ : path_in_C x₁ y₁) (t₁ : ℝ) (
  γ₂ : path_in_C x₂ y₂) (t₂ : ℝ) : ℝ :=
  pathDirection γ₂ t₂ - pathDirection γ₁ t₁
```

```
/--
```

```
The angle between two paths is equal to the argument of the product of the
second path's derivative
and the conjugate of the first path's derivative, modulo 2π.
We prove this equality here assuming the angle difference stays within the
principal branch bounds.
```

```
-/
```

$$\text{Angle}(\gamma_1, t_1, \gamma_2, t_2) = \arg(\gamma_2'(t_2)) - \arg(\gamma_1'(t_1))$$

- `conformal` ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$): f preserves angles between any two paths intersecting at z_0 with non-zero derivatives.

```
def conformal (f : ℂ → ℂ) (z₀ : ℂ) : Prop :=
  ∀ (x₁ y₁ x₂ y₂ : ℂ) (γ₁ : path_in_C x₁ y₁) (t₁ : ℝ) (γ₂ : path_in_C x₂ y₂) (t₂
    : ℝ) (d₁ d₂ : ℂ),
    γ₁.extend t₁ = z₀ →
    γ₂.extend t₂ = z₀ →
    HasDerivAt_R_to_C_eps γ₁.extend (pathDeriv γ₁ t₁) t₁ →
    HasDerivAt_R_to_C_eps γ₂.extend (pathDeriv γ₂ t₂) t₂ →
    pathDeriv γ₁ t₁ ≠ 0 →
    pathDeriv γ₂ t₂ ≠ 0 →
    -- HasDerivAt_R_to_C_eps f f' t says f' is the derivative of f at t.
    HasDerivAt_R_to_C_eps (f ∘ γ₁.extend) d₁ t₁ →
    HasDerivAt_R_to_C_eps (f ∘ γ₂.extend) d₂ t₂ →
    d₁ ≠ 0 ∧ d₂ ≠ 0 ∧ arg d₂ - arg d₁ = pathAngle γ₁ t₁ γ₂ t₂
```

```
/--
```

```
\SII.11 Angle preservation at a point. CONFORMALITY.
```

```
If f is holomorphic at z₀ with non-zero derivative, then it preserves angles
between regular paths intersecting at z₀.
```

```
-/
```

Conformal f at $z_0 \iff f$ preserves angles between all regular intersecting curves at z_0

- `centered_line_path` ($z, v \in \mathbb{C}$): A straight-line path from $z - v$ to $z + v$ evaluating to z at $t = 1/2$.

```
noncomputable def centered_line_path (z v : ℂ) : path_in_ℂ (z - v) (z + v)
  where
  toFun := fun t => z + (2 * (t : ℝ) - 1) * v
```

$$\gamma(t) = z + (2t - 1)v$$

5 ComplexAnalysis/Sarason/Chapter3.lean

- `translationMatrix` ($b \in \mathbb{C}$):

```
noncomputable def translationMatrix (b : ℂ) : GL (Fin 2) ℂ :=
  Units.mk0 !![1, b; 0, 1] ...
```

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

- `scalingMatrix` ($a \in \mathbb{C}, a \neq 0$):

```
noncomputable def scalingMatrix (a : ℂ) (ha : a ≠ 0) : GL (Fin 2) ℂ :=
  Units.mk0 !![a, 0; 0, 1] ...
```

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

- `inversionMatrix`:

```
noncomputable def inversionMatrix : GL (Fin 2) ℂ :=
  Units.mk0 !![0, 1; 1, 0] ...
```

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$